Quiz 4-5

April 11, 2025

Answer all questions. Closed book, closed notes. Time: 40 minutes

1. In 2022, 2 out of 100 students from a coaching centre in Ahmedabad scored ≥ 99 percentile in CAT. Let θ be the probability of scoring ≥ 99 percentile.

   (a) State the continuous Bayes' theorem and its use for the posterior of θ.

   (b) Explain why the posterior updates prior beliefs using data.

   (c) Find the posterior of θ with prior Beta$(0.01, 0.99)$.

   (d) Find the posterior mean and how to get a 99% credible interval.

   (Points $5 + 3 + 8 + 4 = 20$)

2. Let $\theta = \mu$, estimator $\hat{\theta}_n = \bar{x}_n$ for sample size n. Define jackknife and bootstrap variance estimators of $\hat{\theta}_n$. Prove they equal S^2/n for $\hat{\theta}_n = \bar{x}_n$, where S^2 is the sample variance. (Points $6 + 7 + 7 = 20$)

Answers

1. (a) **Bayes' Theorem**: The continuous Bayes' theorem is:

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{\int p(D|\theta)p(\theta)\,d\theta},$$

where $p(D|\theta)$ is the likelihood, $p(\theta)$ the prior, and the denominator normalizes the posterior. For D: 2 successes in 100 trials, the likelihood is binomial, $p(D|\theta) \propto \theta^2(1-\theta)^{98}$. A beta prior simplifies the posterior to another beta distribution via conjugation.

(b) **Updating Beliefs**: The posterior $p(\theta|D)$ updates the prior $p(\theta)$ by incorporating the likelihood $p(D|\theta)$. Here, observing 2 successes in 100 trials adjusts our initial belief about θ (e.g., low probability) toward values consistent with the data, balancing prior and evidence.

(c) **Posterior**: Given $x = 2$ successes, $n = 100$, prior $\theta \sim \text{Beta}(0.01, 0.99)$. Likelihood:

$$p(x|\theta) \propto \theta^2(1-\theta)^{98}.$$

Prior:

$$p(\theta) \propto \theta^{0.01-1}(1-\theta)^{0.99-1}.$$

Posterior:

$$p(\theta|x) \propto \theta^{2+0.01-1}(1-\theta)^{98+0.99-1} = \theta^{2.01-1}(1-\theta)^{98.99-1}.$$

Thus:

$$\theta|x \sim \text{Beta}(2.01, 98.99).$$

(d) **Mean and Interval**: Posterior mean for $\text{Beta}(a, b)$ is $a/(a+b)$:

$$\mathbb{E}[\theta|x] = \frac{2.01}{2.01 + 98.99} = \frac{2.01}{101} \approx 0.0199.$$

For a 99% credible interval, find the 0.5% and 99.5% quantiles of $\text{Beta}(2.01, 98.99)$ using numerical methods (e.g., R's `qbeta(0.005, 2.01, 98.99)`, `qbeta(0.995, 2.01, 98.99)`). This interval contains 99% of the posterior probability.

2. (a) **Definitions**: **Jackknife**: For $\hat{\theta}_n = \bar{x}_n$, compute leave-one-out estimates:

$$\hat{\theta}_{(i)} = \frac{1}{n-1}\sum_{j \neq i} x_j.$$

Variance:

$$\hat{V}_{\text{jack}} = \frac{n-1}{n}\sum_{i=1}^{n}(\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)})^2, \quad \hat{\theta}_{(\cdot)} = \frac{1}{n}\sum_{i=1}^{n}\hat{\theta}_{(i)}.$$

Bootstrap: Draw B samples of size n with replacement, compute $\bar{x}^{*b}$ per sample. Variance:

$$\hat{V}_{\text{boot}} = \frac{1}{B-1}\sum_{b=1}^{B}(\bar{x}^{*b} - \bar{x}^{*})^2, \quad \bar{x}^{*} = \frac{1}{B}\sum_{b=1}^{B}\bar{x}^{*b}.$$

(b) **Jackknife Variance**: For $\bar{x}_n$:

$$\hat{\theta}_{(i)} = \frac{n\bar{x}_n - x_i}{n-1}.$$

$$\hat{\theta}_{(\cdot)} = \frac{1}{n}\sum_{i=1}^{n}\frac{n\bar{x}_n - x_i}{n-1} = \frac{n^2\bar{x}_n - n\bar{x}_n}{n(n-1)} = \bar{x}_n.$$

$$\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)} = \frac{\bar{x}_n - x_i}{n-1}.$$

$$\sum_{i=1}^{n}(\hat{\theta}_{(i)} - \hat{\theta}_{(\cdot)})^2 = \frac{1}{(n-1)^2}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2 = \frac{(n-1)S^2}{(n-1)^2} = \frac{S^2}{n-1}.$$

$$\hat{V}_{\text{jack}} = \frac{n-1}{n}\cdot\frac{S^2}{n-1} = \frac{S^2}{n}.$$

(c) **Bootstrap Variance**: Each $\bar{x}^{*b}$ is the mean of a sample with replacement. Let's write

$\bar{x}^{*b} = n^{-1}\sum_{i=1,2,...,n} n_i^* x_i$

where n_i^* represents the frequency of x_i in the bootstrap sample. It is known $(n_1^*, n_2^*, ..., n_n^*) \backsim$ (follows a multinomial distribution) $\text{Mult}(n; 1/n, 1/n, ..., 1/n)$. Clearly $E(n_i^*) = n \times 1/n = 1$, $Var(n_i^*) = n\times 1/n\times(1-1/n) = (n-1)/n$ and $Covar(n_i^*, n_j^*) = -n\times(1/n)(1/n) = -(1/n)$

Now $Var(\bar{x}^{*b}) = \sum_{i=1,2,...,n} x_i^2 var(n_i^*) + \sum_{i\neq j, i,j=1,2,...,n} Covar(n_i^*, n_j^*)$. After simplification, it reduces to S^2/n.